

Aniketh Janardhan Reddy

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Education

- University of California, Berkeley, Berkeley, CA,** Expected 2026
Doctor of Philosophy (PhD) in Computer Science, GPA: 4.0/4.0.
- Carnegie Mellon University - School of Computer Science, Pittsburgh, PA,** Dec 2019
Master of Science in Machine Learning, QPA: 4.11/4.33.
- Birla Institute of Technology and Science (BITS), Pilani, Hyderabad, India,** Jul 2018
Bachelor of Engineering (Honours) Computer Science, CGPA: 9.93/10.

Experience

- Graduate Student, UC Berkeley, Berkeley, CA.** Jan 2021–Present
- I am advised by Prof. Nilah Ioannidis and we are currently working on building better gene expression prediction models and on optimizing sequences to obtain beneficial gene expression profiles.
- Software Engineer, Microsoft, Bellevue, WA.** Jan 2020–Jan 2021
- Worked on improving the quality of related searches shown on the Bing image search vertical. This was done by both improving ranking algorithms and the quality of candidate suggestions, mostly through the deployment of state-of-the-art ML models at scale. These changes significantly boosted user engagement and revenue.
- Software Engineer Intern, Microsoft, Sunnyvale, CA.** May 2019–Aug 2019
- Devised a new demand-based autoscaler for Application Gateways as a member of the Azure Networking group. This led to a ~66% reduction in the number of dropped requests compared to the previous system.
- Research Assistant, Carnegie Mellon University, Pittsburgh, PA.** Aug 2018–Dec 2019
- Worked with Prof. Leila Wehbe on understanding the representation of language syntax in the human brain using functional magnetic resonance imaging (fMRI) and machine learning.
 - Effort-based metrics widely used to study syntax processing do not explicitly encode syntactic structure. We proposed subgraph embedding-based features that encode syntactic structure and showed that they are predictive of brain activity, even after controlling for effort. Our results indicated that syntax processing is distributed across the language network and that there is some evidence of hierarchical structure building and prediction of future syntactic information in the brain. Using BERT embeddings, we also showed that the regions that process syntax are a subset of those that process semantics.
- Intern, Smart Data Analytics Research, Universität Bonn, Bonn, Germany.** May 2017–Aug 2018
- Worked on several projects in the broad area of automated fact verification under the tutelage of Prof. Dr. Jens Lehmann.
 - Developed a fact verification system called DeFactoNLP which could not only assess the veracity of a claim but also retrieve supporting evidences from Wikipedia. The system made use of named entity recognition (NER), TF-IDF vector comparisons and a Decomposable Attention-based natural language inference model. It achieved a 0.4277 evidence F1-score and a 51.36% label accuracy which were 134% and 5% above the baseline scores respectively.
 - Built a state-of-the-art system which could ascertain the credibility of a website using handcrafted features derived from the text and visuals present on the website. Also, co-developed and benchmarked DeFacto, a triple verification framework.
- Research Intern, Cognition Lab, Indian Institute of Science (IISc), Bangalore, India.** Jan 2018–May 2018
- Employed fMRI and machine learning to identify brain regions involved in covert attention modulation under the guidance of Prof. Sridharan Devarajan.
 - This was accomplished by analyzing which regions' activations were most predictive of times at which a participant was required to pay covert attention. Many deep learning-based seq2seq models were tested in this context.
 - Our analysis showed that certain regions of the frontal lobe, the basal ganglia and the posterior parietal cortex all act in unison to modulate covert attention and that the effects of attention modulation can be seen in visual cortical areas.

Publications

- **Aniketh Janardhan Reddy** and Leila Wehbe. 2021. Can fMRI reveal the representation of syntactic structure in the brain? Advances in Neural Information Processing Systems 34 (NeurIPS 2021). [Paper][Code]
- **Aniketh Janardhan Reddy** and Sridharan Devarajan. 2018. Analyzing Human Attentional Mechanisms using Functional Magnetic Resonance Imaging Data and Machine Learning. *Undergraduate thesis*. [Paper]
- **Aniketh Janardhan Reddy**, Gil Rocha, and Diego Esteves. 2018. DeFactoNLP: Fact Verification using Entity Recognition, TFIDF Vector Comparison and Decomposable Attention. FEVER 2018 at EMNLP 2018. [Paper][Code]
- Diego Esteves, **Aniketh Janardhan Reddy**, Piyush Chawla, and Jens Lehmann. 2018. Belittling the Source: Trustworthiness Indicators to Obfuscate Fake News on the Web. FEVER 2018 at EMNLP 2018. [Paper][Code]
- **Aniketh Janardhan Reddy**, Monica Adusumilli, Sai Kiranmai Gorla, Lalita Bhanu Murthy Neti, and Aruna Malapati. 2018. Named Entity Recognition for Telugu using LSTM-CRF. WILDRE4 at LREC 2018. [Paper][Code]
- Diego Esteves, Anisa Rula, **Aniketh Janardhan Reddy**, and Jens Lehmann. 2018. Toward Veracity Assessment in RDF Knowledge Bases: An Exploratory Analysis. J. Data and Information Quality 9, 3, Article 16 (February 2018). [Paper]

Honors

- Gold Medal**, for best academic record (highest CGPA), BITS Pilani, Hyderabad. Aug 2018
- Best All-Rounder Award**, in recognition of multidimensional achievements, BITS Pilani, Hyderabad. Aug 2018
- Technology Entrepreneurship Program**, Indian School of Business (ISB), Hyderabad, India. Jun 2018
- Merit Scholarship**, BITS Pilani, Hyderabad. 2014–18
- Working Internships in Science and Engineering Scholar**, German Academic Exchange Service. 2017